FedRE: Federated Learning via Representation Entanglement

Anonymous submission

Abstract

Federated learning (FL) enables collaborative learning across clients without compromising privacy. Most existing FL studies focus on model-homogeneous scenarios. However, ensuring uniform model architecture across clients is challenging, leading to a *model-heterogeneous FL* problem. To address this problem, we rethink client knowledge and design a novel form termed the *entangled representation*, which entangles all representations from distinct categories on each client into a single representation. Building on this concept, we propose a Federated Representation Entanglement (FedRE), which first synthesizes **a single entangled representation on each client** and then uploads it to the server to train the global classifier. Each entangled representation integrates information from multiple categories and is categorized into distinct categories according to their respective contributions, enabling the global classifier to learn inter-category decision boundaries. As a result, the entangled representations offer effective, privacy-preserving, and lightweight client knowledge. Theoretically, we analyze the convergence of FedRE, and empirically, we demonstrate its superiority in balancing model performance, privacy protection, and communication overhead.

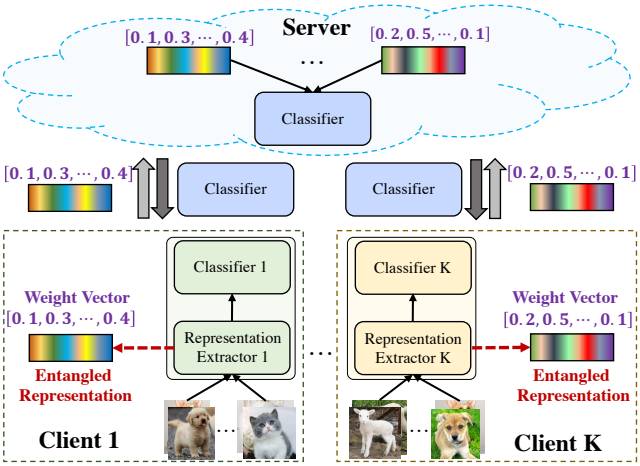


Figure 1: FedRE workflow. Each client has a local model including a representation extractor and a classifier. All representations are entangled into a single representation using a random normalized weight vector on each client, and both are then uploaded to the server to train the global classifier.

Introduction

Federated learning (FL) (McMahan et al. 2017; Yang et al. 2019) is a collaborative learning paradigm that aggregates privacy-preserving client knowledge (*e.g.*, model parameters) from multiple clients. Numerous FL methods have been developed and applied in various fields, such as healthcare (Antunes et al. 2022; Zhang et al. 2025) and the Internet of Things (Nguyen et al. 2021; Fan et al. 2024). Most existing FL studies (McMahan et al. 2017; Zhang et al. 2023b; Li et al. 2023a; Zhang et al. 2024a) assume that the architectures of local models across clients are homogeneous. In practice, however, assuming the same model architecture for all clients is unrealistic due to differences in sample distribution, hardware, and computational capabilities. Also, the model architecture adopted by each client is private and may not be shared with the server or other clients. Those issues motivate a practical but challenging problem known as *model-heterogeneous FL* (Ye et al. 2023), where the representation extractors adopt heterogeneous architectures across clients, while the classifiers share a homogeneous architecture.

To address this problem, various model-heterogeneous FL methods(Li and Wang 2019; Itahara et al. 2021; Makhija et al. 2022; Tan et al. 2022; Liang et al. 2020; Yi et al. 2023; Huang et al. 2023; Wang et al. 2024; Yi et al. 2024; Wu et al. 2024) have been proposed. Most of them focus on aggregating the *representations* (Makhija et al. 2022), *logits* (Itahara et al. 2021), *classifiers* (Liang et al. 2020), *prototypes* (*i.e.*, category means) (Tan et al. 2022; Yi et al. 2023; Huang et al. 2023; Wang et al. 2024), or *small-models* (Yi et al. 2024; Wu et al. 2024), from clients. On the one hand, although representations, logits, and small-models encode high-level client knowledge, uploading them to the server incurs significant communication overhead and exposes a serious privacy risk, as they can be exploited to infer the original samples by launching representation and model inversion attacks (Ulyanov, Vedaldi, and Lempitsky 2018; Yin et al. 2020). On the other hand, uploading classifiers and prototypes that encapsulate representative category-specific knowledge to the server reduces both communication overhead and the risk of inferring original samples. However, reliance on such category-representative knowledge may lead to suboptimal model performance. This raises a question: "*For model-heterogeneous FL, is there more effective, privacy-*

preserving, and lightweight client knowledge available?"

To address this issue, we introduce a novel form of client knowledge, termed **entangled representation**, which incorporates representations from multiple categories into a unified representation. Building on this knowledge, we propose *Federated Representation Entanglement* (FedRE), as illustrated in Figure 1. In FedRE, each client has a local model comprising a representation extractor and a classifier. On each client, all representations are entangled into a single representation using a random normalized weight vector. This entangled representation along with the associated weight vector is then uploaded to the server to train the global classifier. During training, each entangled representation is simultaneously classified into multiple categories based on the corresponding category weights, enabling the global classifier to learn inter-category decision boundaries. Furthermore, the entangled representations can effectively resist representation inversion attacks, as they do not correspond to any single sample, and significantly reduce communication overhead by uploading only a single representation per client. As a result, those entangled representations provide privacy-preserving, lightweight, and effective client knowledge.

The main contributions of this paper are three-fold.

- To the best of our knowledge, we are the first to utilize the entangled representation as client knowledge to train the global classifier.
- We propose FedRE based on the entangled representation and analyze its theoretical convergence.
- Extensive experimental results demonstrate that FedRE achieves an effective balance between model performance, privacy protection, and communication overhead.

Related Work

Existing FL approaches can be roughly categorized into model-heterogeneous and model-homogeneous methods based on their ability to handle model heterogeneity.

Model-heterogeneous FL method handles both heterogeneous local models and sample distributions. Due to the heterogeneity of local models, it is not feasible to aggregate all their parameters. Thus, most of the studies turns to aggregate client knowledge (*e.g.*, representations (Makhija et al. 2022), logits (Itahara et al. 2021), small-models (Yi et al. 2024; Wu et al. 2024), classifiers (Liang et al. 2020), or prototypes (Tan et al. 2022; Yi et al. 2023; Huang et al. 2023; Wang et al. 2024)). For example, DS-FL (Itahara et al. 2021) designs a logit aggregation strategy that integrates the local logits from different clients into the global logit on the server. FedHeNN (Makhija et al. 2022) aligns the representations extracted by distinct local models using a common representation alignment dataset. LG-FedAvg (Liang et al. 2020) aggregates the classifiers from distinct clients on the server. FedProto (Tan et al. 2022), FPL (Huang et al. 2023), FedGH (Yi et al. 2023), and FedTGP (Zhang et al. 2024b) treat prototypes as a form of client knowledge. Specifically, FedProto (Tan et al. 2022) averages the local prototypes of each category to aid in learning local representation extractors. FPL employs a clustering strategy to derive unbiased global prototypes, and FedTGP optimizes trainable global prototypes dynamically. Moreover, FedGH (Yi et al. 2023) utilizes the prototypes from multiple clients to train the global classifier on the server. Additionally, several studies (Zhu, Hong, and Zhou 2021; Wu 2022) focus on knowledge distillation. For instance, FedGen (Zhu, Hong, and Zhou 2021) learns a global generator to augment the training samples for local models, while FedKD (Wu 2022) distills a global student model to assist the learning of local models. Furthermore, another line of research (Wu et al. 2024; Yi et al. 2024) uses homogeneous small-models as client knowledge. A recent example is FedMRL (Yi et al. 2024), which facilitates inter-client knowledge aggregation via a shared small-model.

Model-homogeneous FL method deals with homogeneous local models but heterogeneous sample distributions. Most of the studies (McMahan et al. 2017; Zhang et al. 2023b; Ma et al. 2022; Li et al. 2023a; Deng et al. 2022, 2021; Wang et al. 2021) focus on aggregating all parameters of local models. For instance, FedAvg (McMahan et al. 2017) aggregates all parameters of local models from distinct clients on the server. Based on FedAvg, FedAvgDBE (Zhang et al. 2023a), FedFN (Kim et al. 2023), and FedDecorr (Shi et al. 2023) alleviates the representation bias issue in local models. Another example is FedALA (Zhang et al. 2023b), which adaptively integrates the global and local models to align with the local objective. In addition, several studies (Sun et al. 2023; Collins et al. 2021; Oh, Kim, and Yun 2022; Pillutla et al. 2022) aim to aggregate partial parameters of local models. For example, FedRep (Collins et al. 2021) aggregates the representation extractors of local models to enhance representation capability. Moreover, FedBABU (Oh, Kim, and Yun 2022), SphereFed (Dong et al. 2022), FedETF (Li et al. 2023b), and FedDr+ (Kim et al. 2025) only updates the representation extractors during local training and then aggregates them on the server.

Problem Formulations

In this section, we provide the problem formulations for the model-heterogeneous FL and the threat model.

Model-heterogeneous FL. Given K clients and a server, each client k holds a private local dataset $\mathcal{D}_k = \{(\mathbf{x}_i^k, \mathbf{y}_i^k)\}_{i=1}^{n_k}$, where $\mathbf{x}_i^k$ represents the i-th sample in $\mathcal{D}_k$, and $\mathbf{y}_i^k$ denotes its one-hot label over C categories. The local model at client k is defined as $h_k(\boldsymbol{\theta}_k; \cdot) = \boldsymbol{g}_k(\boldsymbol{\phi}_k; \cdot) \circ f_k(\boldsymbol{\omega}_k; \cdot)$, where $\boldsymbol{g}_k$ with parameter $\boldsymbol{\phi}_k$ acts as the representation extractor, and f_k with parameters $\boldsymbol{\omega}_k$ is the local classifier. There exist distinct clients i and j such that $\boldsymbol{g}_i$ and $\boldsymbol{g}_j$ have heterogeneous architectures, while f_i and f_j share a homogeneous architecture across all clients. The goal is to train client models on $\{\mathcal{D}_k\}_{k=1}^K$ to achieve optimal average classification accuracy across all clients.

Threat Model. We assume a *semi-honest* server that attempts to infer original samples by launching representation inversion attacks (Ulyanov, Vedaldi, and Lempitsky 2018) and can *illegally* access clients' representation extractors.

Methodology

In this section, we introduce the proposed FedRE, detailing its motivation, workflow, and associated analyses.

Motivation

To address the model-heterogeneous FL problem, a promising approach is to incorporate knowledge from multiple clients to train a high-quality global classifier without compromising privacy. This classifier leverages cross-client knowledge and improves local model performance upon deployment to clients. A vanilla method, FedAllRep, uploads all sample **representations** to the server as client knowledge for training the global classifier. While FedAllRep achieves superior model performance, it poses significant risks of leaking original samples through representation inversion attacks (Ulyanov, Vedaldi, and Lempitsky 2018) and incurs substantial communication overhead. To alleviate this issue, FedGH (Yi et al. 2023) uses **prototypes** as client knowledge to train the global classifier, enhancing privacy protection and reducing communication overhead. However, those prototypes, limited to one per category, only encode the representative knowledge of their respective categories. Consequently, training the global classifier on them may lead to overfitting and hinder its ability to learn inter-category decision boundaries.

This limitation motivates the design of **entangled representations**, a novel form of client knowledge that, unlike prototypes, integrates representations from distinct categories. Specifically, we first entangle all representations on each client into a single representation, with each category contributing via a randomly assigned weight. This random weighting introduces variability across global communication rounds, promoting diversity in the entangled representations. Each entangled representation is then uploaded to the server and simultaneously categorized into multiple classes to train the global classifier. Leveraging those cross-category, diverse entangled representations enables the global classifier to better learn inter-category decision boundaries and mitigate the risk of overfitting compared to using prototypes, thereby improving generalization performance. Additionally, the entangled representation provides enhanced privacy protection since it cannot be mapped to any individual sample. Furthermore, uploading only an entangled representation per client significantly reduces communication overhead. In summary, the entangled representations provide effective, privacy-preserving, and lightweight client knowledge, forming the FedRE's foundation. Next, we detail the FedRE.

FedRE

In FedRE, each client has a local model comprising a representation extractor and a classifier, as depicted in Figure 1. The FedRE workflow consists of three main steps: (i) local model update; (ii) representation entanglement and upload; and (iii) global classifier update and broadcast.

Local Model Update Similar to vanilla federated learning approaches such as FedAvg (McMahan et al. 2017), FedRE requires each client to update its local model to effectively learn from local samples. To accomplish this, the optimization objective for the k-th client is formulated by

$$\min_{\boldsymbol{\theta}_k} \frac{1}{n_k} \sum_{(\mathbf{x}_i^k, \mathbf{y}_i^k) \in \mathcal{D}_k} \mathcal{L}_{ce}\big[h_k(\boldsymbol{\theta}_k; \mathbf{x}_i^k), \mathbf{y}_i^k\big], \tag{1}$$

where $\mathcal{L}_{ce}(\cdot, \cdot)$ denotes the cross-entropy loss.

Representation Entanglement and Upload We now introduce how to calculate a single entangled representation on each client. First, to address the heterogeneity of representations across clients with varying model architectures, we apply a simple average pooling $\mathrm{AP}(\cdot)$ (Gholamalinezhad and Khosravi 2020) operation to map all client representations into a unified space, ensuring consistency and enabling the use of a shared global classifier. This approach allows for effective representation aggregation despite architectural differences, facilitating the collaborative learning of heterogeneous models (see Analysis section for an analysis of other representation mapping operations). Then, we calculate the entangled representation as

$$\widetilde{\mathbf{r}}_k = \sum_{c \in \mathcal{C}_k} \frac{w_k^c}{|\mathcal{D}_k^c|} \sum_{(\mathbf{x}_i^k, \mathbf{y}_i^k) \in \mathcal{D}_k^c} \mathrm{AP}\big[\boldsymbol{g}_k(\boldsymbol{\phi}_k; \mathbf{x}_i^k)\big], \tag{2}$$

where $\mathcal{D}_k^c$ be the set of samples belonging to category c in the k-th client, $\mathcal{C}_k$ be the label set of the k-th client, and $\mathbf{w}_k = [w_k^1, \cdots, w_k^{|\mathcal{C}_k|}]$ is a normalized weight vector, where the elements are randomly drawn from a uniform distribution $\mathcal{U}(0, 1)$, and then normalized by dividing each element by the sum of all elements to ensure their sum equals one. Besides this representation entanglement (RE) method, we also detail other methods in the Empirical Analysis section. *The weight vector indicates the probability distribution of each entangled representation across different categories, which can be interpreted as its label encoding*. The specific usage is detailed below.

Global Classifier Update and Broadcast Upon receiving the entangled representations and their corresponding weight vectors $\widetilde{\mathcal{R}} = \{(\widetilde{\mathbf{r}}_k, \mathbf{w}_k)\}_{k=1}^K$, the server utilizes those entangled representations and their associated weight vectors to update the global classifier. Accordingly, the server's optimization objective is formulated as

$$\min_{\boldsymbol{\omega}} \sum_{k=1}^{K} \sum_{c \in \mathcal{C}_k} w_k^c \mathcal{L}_{ce}\big[f(\boldsymbol{\omega}; \widetilde{\mathbf{r}}_k), \mathbf{y}_c\big], \tag{3}$$

where $f(\boldsymbol{\omega}; \cdot)$ represents the global classifier with parameters of $\boldsymbol{\omega}$, $\mathbf{y}_c$ stands for the one-hot label associated with category c. By minimizing Eq. (3), we learn a global classifier based on the entangled representations, which can effectively categorize all training samples from various clients. Finally, the server broadcasts the updated global classifier to all clients for the next iteration.

With the above update process, the FedRE method can be summarized in Algorithm 1.

Analysis

Comparison with Mixup We now compare RE and mixup (Zhang et al. 2018; Verma et al. 2019). Mixup is a data augmentation technique that aims to linearly interpolate two representations and their corresponding labels to synthesize a new representation. Let $\mathcal{R} = \{(\mathbf{r}_i, \mathbf{y}_i)\}_{i=1}^n$ denote the set of representations on a single client, where $\mathbf{r}_i$ represents the representation of the i-th sample, and $\mathbf{y}_i$ denotes its associated one-hot label over C categories. Recall that mixup is

Algorithm 1: FedRE

Input: K clients with their respective datasets $\{\mathcal{D}_k\}_{k=1}^K$.
Output: Local models for all clients, *i.e.*, $\{h_k(\boldsymbol{\theta}_k;\cdot)\}_{k=1}^K$.

```
Randomly initialize the global classifier f(ω;·) and the local models {h_k(θ_k;·)}_{k=1}^K.
for t = 0 to T − 1 do
    for each client k in parallel do                    ▷ Client Side
        Receive ω to update ω_k and θ_k by Eq. (1).
        Randomly generate and normalize w_k.
        Calculate r̃_k by Eq. (2) and upload (r̃_k, w_k) to the server.
    end for
    Update ω according to Eq. (3) and broadcast ω to all clients.   ▷ Server Side
end for
```

formulated as follows:

$$\widetilde{\mathbf{r}}_{\text{mixup}} = \lambda \mathbf{r}_i + (1-\lambda)\mathbf{r}_j, \widetilde{\mathbf{y}}_{\text{mixup}} = \lambda \mathbf{y}_i + (1-\lambda)\mathbf{y}_j, \quad (4)$$

where $\lambda \sim \text{Beta}(\alpha, \alpha)$, for $\alpha \in (0, \infty)$. In contrast, RE, specifically designed for FL, aims to ensure model performance while reducing communication costs and privacy risks. Its general form is formulated as follows:

$$\widetilde{\mathbf{r}}_{\text{RE}} = \sum_{i=1}^{n} w_i \mathbf{r}_i, \widetilde{\mathbf{y}}_{\text{RE}} = \sum_{i=1}^{n} w_i \mathbf{y}_i, \quad (5)$$

where $w_i \in [0, 1]$ is the weight of $\mathbf{r}_i$ and can be determined by various RE methods (see details in Appendix A). As indicated by Eqs. (4)–(5), mixup performs linear interpolation between pairs of representations, while RE entangles the entire set of representations from each client using a weighted sum. Furthermore, RE trains the classifier by calculating a weighted sum of losses across multiple categories, with the weights determined by their contributions.

Computational Complexity We analyze the computational complexity of RE on the client side. Let $\mathbf{R} \in \mathbb{R}^{n \times d}$ be the representation matrix, where n is the number of samples and d is the dimensionality. Let $\mathbf{Y} \in \mathbb{R}^{n \times C}$ be the corresponding one-hot label matrix of $\mathbf{R}$, where C is the number of categories, and let $\mathbf{w} \in \mathbb{R}^C$ be the weight vector. The entangled representation can be calculated as $\widetilde{\mathbf{r}} = \mathbf{R}^\top(\mathbf{Y}\mathbf{w})$, resulting in a computational complexity of $\mathcal{O}\big(n \times (d + C)\big)$.

Convergence We present the convergence analysis of FedRE in the model-heterogeneous setting, focusing on the convergence conditions and rate for the local objective of an arbitrary client. Let $t \in \{0, \ldots, T-1\}$ be the global communication round and $e \in \{0, 1, \ldots, E\}$ be the local iteration. At the start of the $(t+1)$-th round, denoted as $(tE+0)$, each client replaces the local classifier with the global one trained in the t-th round. The first iteration of the $(t+1)$-th round is $(tE+1)$, and $(tE+E)$ marks the final iteration of that round. Our convergence analysis is based on the following assumptions, which are similar to those used in existing FL studies (Yi et al. 2023; Tan et al. 2022).

Assumption 1. *Each local objective function is L_1-Lipschitz smooth, i.e., there exists a constant $L_1 > 0$ such that $\|\nabla\mathcal{L}\left(\boldsymbol{\theta}_k^{t_1};\mathcal{D}_k\right) - \nabla\mathcal{L}\left(\boldsymbol{\theta}_k^{t_2};\mathcal{D}_k\right)\|_2 \leq L_1 \left\|\boldsymbol{\theta}_k^{t_1} - \boldsymbol{\theta}_k^{t_2}\right\|_2$, $\forall t_1, t_2 > 0$, $k \in \{1, \cdots, K\}$.*

Assumption 2. *The stochastic gradient on each client is unbiased, i.e., $\mathbb{E}_{\mathcal{B}_k \subseteq \mathcal{D}_k}\left[\nabla\mathcal{L}\left(\boldsymbol{\theta}_k;\mathcal{B}_k\right)\right] = \nabla\mathcal{L}\left(\boldsymbol{\theta}_k;\mathcal{D}_k\right)$, and its variance is bounded by a constant σ^2, i.e., $\mathbb{E}_{\mathcal{B}_k \subseteq \mathcal{D}_k}\left[\left\|\nabla\mathcal{L}\left(\boldsymbol{\theta}_k;\mathcal{B}_k\right) - \nabla\mathcal{L}\left(\boldsymbol{\theta}_k;\mathcal{D}_k\right)\right\|_2^2\right] \leq \sigma^2$, $k \in \{1, \cdots, K\}$.*

Assumption 3. *The divergence between the local and global classifiers is bounded by two constants ϵ^2 and δ^2, i.e., $\|\boldsymbol{\omega}_k^{t+1} - \boldsymbol{\omega}^t\|_2^2 \leq \epsilon^2$, $\|\boldsymbol{\omega}_k^t - \boldsymbol{\omega}^t\|_2^2 \leq \delta^2$, $\forall t > 0$, $k \in \{1, \cdots, K\}$.*

Based on them, we first establish the convergence conditions for FedRE in Theorem 1 and then derive its convergence rate in Theorem 2.

Theorem 1. *Under Assumptions 1 to 3 and the local learning rate satisfying $\eta_l^{e'} < \frac{2\sum_{e=0}^{e'}\|\nabla L_{tE+e}\|_2^2}{L_1 E\sigma^2 + L_1\sum_{e=0}^{e'}\|\nabla L_{tE+e}\|_2^2}$, $e' \in \{0, 1, \ldots E-1\}$, the local objective functions convergence.*

Theorem 2. *Under Assumptions 1 to 3, let $\Delta = \mathcal{L}_0 - \mathcal{L}^*$ where $\mathcal{L}_0$ and $\mathcal{L}^*$ denote the initial and optimal values of the local objective function, respectively. For an arbitrary client, given any $\xi > 0$, when $T > \frac{2\Delta}{\xi E(2\eta_l - L_1\eta_l^2) - L_1 E\eta_l^2\sigma^2 - L_1\delta^2}$, and the local learning rate satisfying $\eta_l < \frac{2\xi}{\xi L_1 + L_1\sigma^2}$, we have $\frac{1}{TE}\sum_{t=0}^{T-1}\sum_{e=0}^{E-1}\mathbb{E}\left\|\nabla L_{tE+e}\right\|_2^2 \leq \xi$.*

The proofs of Theorems 1 and 2 are in Appendix B.

Experiments

In this section, we evaluate the effectiveness of the FedRE.

Experimental Setup

Datasets and Baselines We use three datasets: CIFAR-10 (Krizhevsky, Hinton et al. 2009), CIFAR-100 (Krizhevsky, Hinton et al. 2009), and TinyImageNet (Le and Yang 2015). Moreover, we compare FedRE with eight state-of-the-art approaches: LG-FedAvg (Liang et al. 2020), FedGH (Yi et al. 2023), FedKD (Wu 2022), FedGen (Zhu, Hong, and Zhou 2021), FedProto (Tan et al. 2022), FPL (Huang et al. 2023), FedMRL (Yi et al. 2024), FedTGP (Zhang et al. 2024b), as well as a Local method that trains local models independently on each client without communication.

Evaluation Metrics We evaluate model performance by calculating the classification accuracy on the test set across all clients. To ensure a fair comparison, we report the average classification accuracy of the final round after 100 rounds, calculated across three random experiments, along with its standard deviation. Communication overhead is measured by the total number of parameters uploaded and broadcast per round, respectively. To evaluate privacy protection, we adopt Peak Signal-to-Noise Ratio (PSNR) (Schlett et al. 2022) and Mean Squared Error (MSE), which measure the fidelity of reconstructed images. Low PSNR and high MSE indicate

Table 1: Accuracy (%) comparison on three datasets under the model-heterogeneous setting. In each column, the best results are **bolded**, and the second-best results are underlined.

Method	PRA			PAT			Average
	CIFAR-10	CIFAR-100	TinyImageNet	CIFAR-10	CIFAR-100	TinyImageNet	
LG-FedAvg (Liang et al. 2020)	80.90 ± 0.17	41.96 ± 0.03	25.16 ± 0.42	85.35 ± 0.25	58.24 ± 0.33	32.26 ± 0.28	53.98
FedGH (Yi et al. 2023)	78.66 ± 0.34	40.91 ± 0.26	25.04 ± 0.11	85.43 ± 0.03	58.07 ± 0.33	31.98 ± 0.29	53.35
FedKD (Wu 2022)	80.79 ± 0.38	41.33 ± 0.25	25.39 ± 0.36	84.03 ± 0.17	55.61 ± 0.10	31.73 ± 0.20	53.15
FedGen (Zhu, Hong, and Zhou 2021)	81.16 ± 0.12	41.46 ± 0.10	25.45 ± 0.19	84.88 ± 0.18	57.87 ± 0.67	31.96 ± 0.21	53.80
FedProto (Tan et al. 2022)	78.36 ± 0.52	35.00 ± 0.34	18.16 ± 0.08	83.81 ± 0.18	56.72 ± 0.11	29.61 ± 0.02	50.28
FPL (Huang et al. 2023)	77.40 ± 0.23	36.66 ± 0.45	22.64 ± 0.46	83.89 ± 0.38	53.21 ± 0.25	29.16 ± 0.19	50.49
FedMRL (Yi et al. 2024)	81.28 ± 0.05	34.41 ± 0.04	20.92 ± 0.09	83.30 ± 0.41	54.25 ± 0.21	27.37 ± 0.10	50.26
FedTGP (Zhang et al. 2024b)	81.32 ± 0.47	35.89 ± 0.22	28.70 ± 0.49	84.68 ± 0.27	54.67 ± 1.34	35.64 ± 0.37	53.48
Local	81.20 ± 0.05	41.57 ± 0.10	25.81 ± 0.15	84.68 ± 0.07	57.96 ± 0.12	33.02 ± 0.14	54.04
FedRE	**82.60 ± 0.01**	**46.36 ± 0.09**	**30.48 ± 0.13**	**86.20 ± 0.14**	**62.56 ± 0.32**	**38.52 ± 0.08**	**57.79**

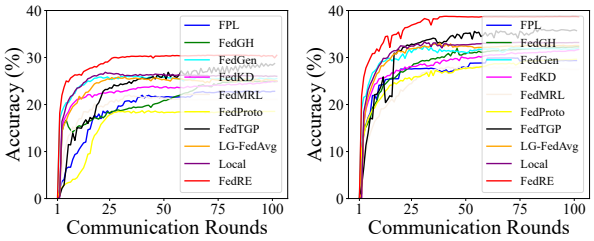
substantial reconstruction errors, suggesting that the original image is difficult to recover and thus privacy is well preserved.

Model-Heterogeneous Settings We configure 10 clients with 10 distinct model architectures: a 4-layer CNN from (Zhang et al. 2023a), MobileNetV2 (Sandler et al. 2018), GoogleNet (Szegedy et al. 2015), five ResNet models (ResNet-18, ResNet-34, ResNet-50, ResNet-101, ResNet-152) (He et al. 2016), and two Vision Transformer (ViT) models (ViT-B/16 and ViT-B/32) (Han et al. 2022).

Statistic-Heterogeneous Settings We follow (Zhang et al. 2023a) to adopt both practical (PRA) (Lin et al. 2020; Li, He, and Song 2021) and pathological (PAT) (Shamsian et al. 2021) settings. In the PRA setting, samples are distributed across clients using a Dirichlet distribution (Lin et al. 2020) with a parameter α, which is set to 0.1 by default across all datasets. In the PAT setting, each client is assigned samples from 2, 10, and 20 categories, drawn from a total of 10, 100, and 200 categories in CIFAR-10, CIFAR-100, and TinyImageNet, respectively, with varying sample sizes. More experimental details can be found in Appendix C.

Main Experiments

Q1: How does FedRE perform in model-heterogeneous settings? The results under the model-heterogeneous FL setting are listed in Table 1. We have several insightful observations. (1) FedRE substantially outperforms all the baselines across various scenarios. In particular, FedRE achieves an accuracy on TinyImageNet that surpasses those of LG-FedAvg, FedGH, and FedKD by 6.26%, 6.54%, and 6.79%, respectively, under the PAT setting. Also, several methods do not exceed the performance of Local, indicating that the scenarios are challenging. (2) FedRE outperforms FedGH, suggesting that using entangled representations to train the global classifier is more effective than using prototypes. One reason is that entangled representations enable the global classifier to learn inter-category decision boundaries more effectively than prototypes. (3) LG-FedAvg is worse than FedRE, which indicates that directly aggregating local classifiers from clients to form the global classifier is less effective than carefully optimizing the global classifier using entangled



(a) PRA (b) PAT

Figure 2: Accuracy (%) comparison between distinct communication rounds on the TinyImageNet dataset in the model-heterogeneous setting.

representations. Also, Figures 2(a)-(b) show that the accuracy of FedRE increases initially and then stabilizes on the TinyImageNet dataset, consistently outperforming other baselines during training, suggesting stable performance convergence. Moreover, we evaluate FedRE under various statistical-heterogeneous settings in **Q5** of Analysis section, including large-scale participation (*i.e.*, **100 clients**).

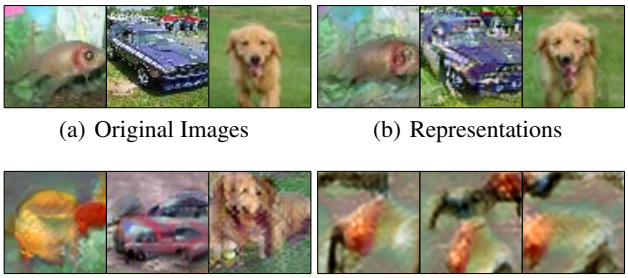
Q2: What is the communication overhead of FedRE? We conduct communication overhead experiments on the CIFAR-100 dataset in the model-heterogeneous settings. As shown in Table 2, FedRE achieves the lowest communication overhead during the upload phase, as it uploads only a single entangled representation from each client to the server. During the broadcast phase, its overhead is comparable to that of classifier-based methods (e.g., LG-FedAvg) and prototype-based methods (e.g., FedProto). More results are offered in Appendix D.1.

Q3: Can entangled representations effectively resist privacy attacks? Following the Threat Model, we reconstruct original samples from representations, prototypes, and entangled representations, respectively. Figure 3 depicts the reconstruction results on TinyImageNet (More results are offered in Appendix D.2). We can make several key observations. (1) Most image contours are reconstructed from the representations, indicating their vulnerability to privacy at-

Table 2: Communication overhead (# params $\times 10^3$) comparison on the CIFAR-100 dataset. In each row, the best results are **bolded**, and the second-best results are underlined.

Metric	LG-FedAvg	FedGH	FedKD	FedGen	FedProto	FedMRL	FedTGP	FPL	FedRE
Upload	513.00	257.02	4234.28	9247.08	257.02	8863.08	257.02	257.02	**5.12**
Broadcast	513.00	**512.00**	4234.28	513.00	**512.00**	8863.08	**512.00**	916.48	513.00

tacks. (2) Some category information, such as the presence of a *fish*, is leaked through reconstructed prototypes, as prototypes encapsulate representative category information. (3) The reconstructed images from entangled representations reveal no identifiable information, demonstrating their effectiveness against privacy attacks. In addition, the PSNR values for images reconstructed from representations, prototypes, and entangled representations are 12.89, 10.25, and 9.66, with corresponding MSE values of 4514.91, 6992.04, and 7781.87. These results indicate that entangled representations yield the lowest PSNR and highest MSE, suggesting stronger privacy protection. Overall, all results demonstrate the superior privacy protection provided by entangled representations.



(a) Original Images (b) Representations

(c) Prototypes (d) Entangled Representations

Figure 3: Comparison of privacy protection in restructuring results from representations, prototypes, and entangled representations on the TinyImageNet dataset.

Empirical Analysis

Q4: What are the advantages of uploading a single entangled representation per client compared to uploading all representations? To explore the benefits of entangled representations, we compare FedRE with FedAllRep, which uses all clients' representations to train the global classifier. Table 3 lists the results on the TinyImageNet dataset in both the PRA and PAT settings. As can be seen, FedRE achieves performance comparable to FedAllRep, suggesting that uploading a single entangled representation per client can effectively support global classifier training. Also, FedRE significantly reduces communication overhead, providing a clear advantage over uploading all representations.

Q5: How does FedRE perform under different participation ratios with varying statistical heterogeneity? We conduct experiments on the CIFAR-10 dataset under the *partial participation setting* with varying levels of statistical heterogeneity. Specifically, we adopt **100 clients** and set participation rates to 10/100 and 20/100, while adjusting the Dirichlet distribution parameter α to 0.07 and 0.1, respectively, in the PRA setting. Furthermore, we follow (Lee et al. 2019) and simulate the long-tail settings by modifying imbalance factors (IF) to 100 and 50, then set α to 0.07. The results in Table 4 show that FedRE outperforms other methods across most scenarios, confirming its effectiveness under partial participation with highly heterogeneous distributions. More results are provided in Appendix D.4.

Table 3: Accuracy (%) and communication overhead comparison on the TinyImageNet dataset. In each column, the best results are **bolded**, and the second-best results are underlined.

Method	PRA		PAT	
	Accuracy	# Params ($\times 10^3$)	Accuracy	# Params ($\times 10^3$)
FedAllRep	**31.20**	42160.39	**38.62**	42258.88
FedRE	30.48	**4118.48**	38.52	**4118.48**

Q6: How effective are various RE methods? We provide a comprehensive analysis of various RE methods (see mathematical details in Appendix A): (1) *Random Select Representation* (**RSR**) randomly selects one representation from each client; (2) *Vanilla Average Representation* (**VAR**) averages all representations per client into a single representation, with equal weight assigned to each; (3) *Random Average Representation* (**RAR**) entangles all representations per client into a single representation using a normalized weight vector, with elements randomly drawn from $\mathcal{U}(0, 1)$ and normalized to sum to one; (4) *Random Select Prototype* (**RSP**) calculates prototypes for each client and randomly selects one prototype per client; (5) *Vanilla Average Prototype* (**VAP**) calculates prototypes for each client and averages them into a single representation, with equal weight assigned to each; (6) *Random Average Prototype* (**RAP**) calculates prototypes for each client and entangles them into a single representation using a normalized weight vector, with elements randomly drawn from $\mathcal{U}(0, 1)$ and normalized to sum to one. Table 5 lists the results on the CIFAR-10 and CIFAR-100 datasets in the PRA setting. We have the following observations. (1) RSR performs the worst, as each client uploads only a randomly selected representation, which is insufficient to train the global classifier. (2) RSP outperforms RSR, as the prototype encodes the knowledge of all representations within a category, it is more representative than a single representation. (3) VAP and RAP outperform VAR and RAR, respectively, indicating that prototype-based entanglement yields better model performance. (4) RAP surpasses VAP, demonstrating that random weights for entanglement are more effective

Table 4: Accuracy (%) comparison for partial participation scenarios with varying statistical heterogeneity in the PRA setting on the CIFAR-10 dataset. Here, α is a Dirichlet distribution parameter, and IF denotes imbalance factors of the long-tail setting. In each column, the best results are **bolded**, and the second-best results are underlined.

Method	Participation rate	$\alpha = 0.07$	$\alpha = 0.1$	IF = 100, $\alpha = 0.07$	IF = 50, $\alpha = 0.07$	Participation rate	$\alpha = 0.07$	$\alpha = 0.1$	IF = 100, $\alpha = 0.07$	IF = 50, $\alpha = 0.07$
FedProto	10 / 100	54.00	51.18	45.21	43.92	20 / 100	56.90	55.47	47.08	44.68
FedGH		78.23	76.87	**67.30**	63.73		80.57	77.84	65.56	64.90
FedRE		**81.17**	**79.56**	67.12	**66.37**		**82.80**	**81.99**	**69.33**	**68.81**

than equal weights. Thus, we empirically choose RAP in the implementation of FedRE.

Table 5: Accuracy (%) comparison between distinct representation entanglement mechanisms on the CIFAR-10 and CIFAR-100 datasets in the PRA setting. In each row, the best results are **bolded**, and the second-best results are underlined.

Dataset	RSR	VAR	RAR	RSP	VAP	RAP
CIFAR-10	79.10	81.32	80.20	80.45	81.42	**82.60**
CIFAR-100	40.41	44.88	43.19	43.25	46.12	**46.36**

Q7: How effective are various representation mapping operations? We analyze various representation mapping operations to address the heterogeneity of representations with different model architectures: (1) Average Pooling (**AP**) maps the representations to a unified space by averaging the values across regions. (2) Max Pooling (**MP**) maps the representations to a unified space by selecting the maximum values across regions. (3) Full Connection (**FC**) maps the representations to a unified space by using a fully connected layer transformation. Table 6 presents the results on the CIFAR-100 dataset in both PRA and PAT settings. We observe that AP achieves the best performance. One possible reason is that AP captures more comprehensive information by averaging all values across regions. Thus, we empirically select AP in the implementation of FedRE.

Table 6: Accuracy (%) comparison between distinct representation mapping operation on the CIFAR-100 dataset in PRA and PAT settings. In each row, the best results are **bolded**, and the second-best results are underlined.

Setting	AP	MP	FC
PRA	**46.36**	45.97	44.53
PAT	**62.56**	61.93	60.19

Q8: How effective are different combinations of representation entanglement and mapping strategies? We further investigate the combination of distinct representation entanglement methods and representation mapping operations. Table 7 lists the results on CIFAR-10 in the model-heterogeneous setting under the PRA setting. Among the tested combinations, RAP + AP yields the highest accuracy compared to other combinations, highlighting the effectiveness of this combination.

Table 7: Accuracy (%) comparison with distinct combinations of representation entanglement and mapping strategies on the CIFAR-10 datasets in the PRA setting. The best results are **bolded**, and the second-best results are underlined.

	RSR	VAR	RAR	RSP	VAP	RAP
AP	79.10	81.32	80.20	80.45	81.42	**82.60**
MP	78.37	81.17	80.31	80.07	80.64	81.93
FC	78.02	80.60	79.69	79.92	80.29	81.28

Q9: Is FedRE still effective in the model-homogeneous setting? We evaluate FedRE under the model-homogeneous setting in both the PRA and PAT settings. In addition to model-heterogeneous methods, we include additional model-homogeneous methods, *i.e.*, FedAvg (McMahan et al. 2017), FedAvgDBE (Zhang et al. 2023a), and FedALA (Zhang et al. 2023b), for comparison. The results are presented in Table 10 in Appendix D.3. As shown, FedRE achieves the best performance across all datasets. Also, Figure 5 in Appendix D.3 shows that FedRE maintains superior accuracy and stable convergence during the training process on the TinyImageNet dataset. Those results confirm the effectiveness and broad applicability of FedRE.

Conclusion

In this paper, we introduce the entangled representation, which serves as an effective, privacy-preserving, and lightweight form of client knowledge. Building on this knowledge, we propose FedRE to address the model-heterogeneous FL problem, where each client entangles all local representations from distinct categories into a single representation and uploads it to the server to collaboratively train the global classifier. We provide a theoretical analysis of FedRE's convergence properties. Experiments on three datasets confirm that FedRE achieves a well-balanced trade-off between model performance, privacy protection, and communication overhead. Accordingly, we believe that the concept of entangled representations offers a novel perspective to balancing those critical factors in FL. One promising direction is to investigate how entangled representations can be adapted to broader machine learning scenarios.

References

Antunes, R. S.; André da Costa, C.; Küderle, A.; Yari, I. A.; and Eskofier, B. 2022. Federated learning for healthcare: Systematic review and architecture proposal. *ACM TIST*, 13(4): 1–23.

Collins, L.; Hassani, H.; Mokhtari, A.; and Shakkottai, S. 2021. Exploiting shared representations for personalized federated learning. In *ICML*, 2089–2099.

Deng, Y.; Lyu, F.; Ren, J.; Chen, Y.-C.; Yang, P.; Zhou, Y.; and Zhang, Y. 2021. Fair: Quality-aware federated learning with precise user incentive and model aggregation. In *INFORCOM*, 1–10.

Deng, Y.; Lyu, F.; Ren, J.; Chen, Y.-C.; Yang, P.; Zhou, Y.; and Zhang, Y. 2022. Improving federated learning with quality-aware user incentive and auto-weighted model aggregation. *TPDS*, 33(12): 4515–4529.

Dong, X.; Zhang, S. Q.; Li, A.; and Kung, H. 2022. Spherefed: Hyperspherical federated learning. In *ECCV*, 165–184.

Fan, J.; Wu, K.; Tang, G.; Zhou, Y.; and Huang, S. 2024. Taking advantage of the mistakes: Rethinking clustered federated learning for iot anomaly detection. *TPDS*.

Gholamalinezhad, H.; and Khosravi, H. 2020. Pooling methods in deep neural networks, a review. *arXiv preprint arXiv:2009.07485*.

Han, K.; Wang, Y.; Chen, H.; Chen, X.; Guo, J.; Liu, Z.; Tang, Y.; Xiao, A.; Xu, C.; Xu, Y.; et al. 2022. A survey on vision transformer. *TPAMI*, 45(1): 87–110.

He, K.; Zhang, X.; Ren, S.; and Sun, J. 2016. Deep residual learning for image recognition. In *CVPR*, 770–778.

Huang, W.; Ye, M.; Shi, Z.; Li, H.; and Du, B. 2023. Rethinking federated learning with domain shift: A prototype view. In *CVPR*, 16312–16322.

Itahara, S.; Nishio, T.; Koda, Y.; Morikura, M.; and Yamamoto, K. 2021. Distillation-based semi-supervised federated learning for communication-efficient collaborative training with non-iid private data. *TMC*, 22(1): 191–205.

Kim, S.; Jeong, M.; Kim, S.; Cho, S.; Ahn, S.; and Yun, S. Y. 2025. FedDr+: Stabilizing Dot-regression with Global Feature Distillation for Federated Learning. *TMLR*.

Kim, S.; Lee, G.; Oh, J.; and Yun, S. Y. 2023. FedFN: Feature Normalization for Alleviating Data Heterogeneity Problem in Federated Learning. In *NeurIPS 2023 Workshop: Federated Learning in the Age of Foundation Models*.

Krizhevsky, A.; Hinton, G.; et al. 2009. Learning multiple layers of features from tiny images. *Technical report, University of Toronto*.

Le, Y.; and Yang, X. 2015. Tiny imagenet visual recognition challenge. *CS 231N*, 7(7): 3.

Lee, K.; Kim, H.; Lee, K.; Suh, C.; and Ramchandran, K. 2019. Synthesizing differentially private datasets using random mixing. In *ISIT*, 542–546.

Li, D.; and Wang, J. 2019. Fedmd: Heterogenous federated learning via model distillation. *arXiv preprint arXiv:1910.03581*.

Li, Q.; He, B.; and Song, D. 2021. Model-contrastive federated learning. In *CVPR*, 10713–10722.

Li, Z.; Lin, T.; Shang, X.; and Wu, C. 2023a. Revisiting weighted aggregation in federated learning with neural networks. In *ICML*, 19767–19788.

Li, Z.; Shang, X.; He, R.; Lin, T.; and Wu, C. 2023b. No fear of classifier biases: Neural collapse inspired federated learning with synthetic and fixed classifier. In *ICCV*, 5319–5329.

Liang, P. P.; Liu, T.; Ziyin, L.; Allen, N. B.; Auerbach, R. P.; Brent, D.; Salakhutdinov, R.; and Morency, L.-P. 2020. Think locally, act globally: Federated learning with local and global representations. *arXiv preprint arXiv:2001.01523*.

Lin, T.; Kong, L.; Stich, S. U.; and Jaggi, M. 2020. Ensemble distillation for robust model fusion in federated learning. In *NeurIPS*, volume 33, 2351–2363.

Ma, X.; Zhang, J.; Guo, S.; and Xu, W. 2022. Layer-wised model aggregation for personalized federated learning. In *CVPR*, 10092–10101.

Makhija, D.; Han, X.; Ho, N.; and Ghosh, J. 2022. Architecture agnostic federated learning for neural networks. In *ICML*, 14860–14870.

McMahan, B.; Moore, E.; Ramage, D.; Hampson, S.; and y Arcas, B. A. 2017. Communication-efficient learning of deep networks from decentralized data. In *AISTATS*, 1273–1282.

Nguyen, D. C.; Ding, M.; Pathirana, P. N.; Seneviratne, A.; Li, J.; and Poor, H. V. 2021. Federated learning for internet of things: A comprehensive survey. *IEEE Commun Surv Tutorials*, 23(3): 1622–1658.

Oh, J.; Kim, S.; and Yun, S.-Y. 2022. FedBABU: Toward Enhanced Representation for Federated Image Classification. In *ICLR*.

Pillutla, K.; Malik, K.; Mohamed, A.-R.; Rabbat, M.; Sanjabi, M.; and Xiao, L. 2022. Federated learning with partial model personalization. In *ICML*, 17716–17758.

Sandler, M.; Howard, A.; Zhu, M.; Zhmoginov, A.; and Chen, L.-C. 2018. Mobilenetv2: Inverted residuals and linear bottlenecks. In *CVPR*, 4510–4520.

Schlett, T.; Rathgeb, C.; Henniger, O.; Galbally, J.; Fierrez, J.; and Busch, C. 2022. Face image quality assessment: A literature survey. *ACM Comput Surv*, 54(10s): 1–49.

Shamsian, A.; Navon, A.; Fetaya, E.; and Chechik, G. 2021. Personalized federated learning using hypernetworks. In *ICML*, 9489–9502.

Shi, Y.; Liang, J.; Zhang, W.; Xue, C.; Tan, V. Y.; and Bai, S. 2023. Understanding and mitigating dimensional collapse in federated learning. *TPAMI*, 46(5): 2936–2949.

Sun, G.; Mendieta, M.; Luo, J.; Wu, S.; and Chen, C. 2023. Fedperfix: Towards partial model personalization of vision transformers in federated learning. In *ICCV*, 4988–4998.

Szegedy, C.; Liu, W.; Jia, Y.; Sermanet, P.; Reed, S.; Anguelov, D.; Erhan, D.; Vanhoucke, V.; and Rabinovich, A. 2015. Going deeper with convolutions. In *CVPR*, 1–9.

Tan, Y.; Long, G.; Liu, L.; Zhou, T.; Lu, Q.; Jiang, J.; and Zhang, C. 2022. Fedproto: Federated prototype learning across heterogeneous clients. In *AAAI*, 8432–8440.

Ulyanov, D.; Vedaldi, A.; and Lempitsky, V. 2018. Deep image prior. In *CVPR*, 9446–9454.

Verma, V.; Lamb, A.; Beckham, C.; Najafi, A.; Mitliagkas, I.; Lopez-Paz, D.; and Bengio, Y. 2019. Manifold mixup: Better representations by interpolating hidden states. In *ICML*, 6438–6447.

Wang, L.; Bian, J.; Zhang, L.; Chen, C.; and Xu, J. 2024. Taming Cross-Domain Representation Variance in Federated Prototype Learning with Heterogeneous Data Domains. *arXiv preprint arXiv:2403.09048*.

Wang, L.; Xu, S.; Wang, X.; and Zhu, Q. 2021. Addressing class imbalance in federated learning. In *AAAI*, volume 35, 10165–10173.

Wu, F.; Wang, X.; Wang, Y.; Liu, T.; Su, L.; and Gao, J. 2024. FIARSE: Model-Heterogeneous Federated Learning via Importance-Aware Submodel Extraction.

Wu, W. F. L. L. e. a., C. 2022. Communication-efficient federated learning via knowledge distillation. *Nat Commun*.

Yang, Q.; Liu, Y.; Chen, T.; and Tong, Y. 2019. Federated machine learning: Concept and applications. *ACM TIST*, 10(2): 1–19.

Ye, M.; Fang, X.; Du, B.; Yuen, P. C.; and Tao, D. 2023. Heterogeneous federated learning: State-of-the-art and research challenges. *ACM Comput Surv*, 56(3): 1–44.

Yi, L.; Wang, G.; Liu, X.; Shi, Z.; and Yu, H. 2023. FedGH: Heterogeneous federated learning with generalized global header. In *ACM MM*, 8686–8696.

Yi, L.; Yu, H.; Ren, C.; Wang, G.; Li, X.; et al. 2024. Federated model heterogeneous matryoshka representation learning. *NeurIPS*, 37: 66431–66454.

Yin, H.; Molchanov, P.; Alvarez, J. M.; Li, Z.; Mallya, A.; Hoiem, D.; Jha, N. K.; and Kautz, J. 2020. Dreaming to distill: Data-free knowledge transfer via deepinversion. In *CVPR*, 8715–8724.

Zhang, F.; Zhai, D.; Bai, G.; Jiang, J.; Ye, Q.; Ji, X.; and Liu, X. 2025. Towards fairness-aware and privacy-preserving enhanced collaborative learning for healthcare. *Nat Commun*, 16(1): 2852.

Zhang, H.; Cisse, M.; Dauphin, Y. N.; and Lopez-Paz, D. 2018. mixup: Beyond empirical risk minimization. In *ICLR*.

Zhang, H.; Li, C.; Kan, N.; Zheng, Z.; Dai, W.; Zou, J.; and Xiong, H. 2024a. Improving Generalization in Federated Learning with Model-Data Mutual Information Regularization: A Posterior Inference Approach. *NeurIPS*, 37: 136646–136678.

Zhang, J.; Hua, Y.; Cao, J.; Wang, H.; Song, T.; XUE, Z.; Ma, R.; and Guan, H. 2023a. Eliminating Domain Bias for Federated Learning in Representation Space. In *NeurIPS*.

Zhang, J.; Hua, Y.; Wang, H.; Song, T.; Xue, Z.; Ma, R.; and Guan, H. 2023b. Fedala: Adaptive local aggregation for personalized federated learning. In *AAAI*, volume 37, 11237–11244.

Zhang, J.; Liu, Y.; Hua, Y.; and Cao, J. 2024b. Fedtgp: Trainable global prototypes with adaptive-margin-enhanced contrastive learning for data and model heterogeneity in federated learning. In *AAAI*, 15, 16768–16776.

Zhang, J.; Liu, Y.; Hua, Y.; Wang, H.; Song, T.; Xue, Z.; Ma, R.; and Cao, J. 2023c. PFLlib: Personalized Federated Learning Algorithm Library. *arXiv preprint arXiv:2312.04992*.

Zhu, Z.; Hong, J.; and Zhou, J. 2021. Data-free knowledge distillation for heterogeneous federated learning. In *ICML*, 12878–12889.

Reproducibility Checklist

1. General Paper Structure

1.1. Includes a conceptual outline and/or pseudocode description of AI methods introduced (yes/partial/no/NA) yes

1.2. Clearly delineates statements that are opinions, hypothesis, and speculation from objective facts and results (yes/no) yes

1.3. Provides well-marked pedagogical references for less-familiar readers to gain background necessary to replicate the paper (yes/no) yes

2. Theoretical Contributions

2.1. Does this paper make theoretical contributions? (yes/no) yes

If yes, please address the following points:

2.2. All assumptions and restrictions are stated clearly and formally (yes/partial/no) yes

2.3. All novel claims are stated formally (e.g., in theorem statements) (yes/partial/no) yes

2.4. Proofs of all novel claims are included (yes/partial/no) yes

2.5. Proof sketches or intuitions are given for complex and/or novel results (yes/partial/no) yes

2.6. Appropriate citations to theoretical tools used are given (yes/partial/no) yes

2.7. All theoretical claims are demonstrated empirically to hold (yes/partial/no/NA) yes

2.8. All experimental code used to eliminate or disprove claims is included (yes/no/NA) yes

3. Dataset Usage

3.1. Does this paper rely on one or more datasets? (yes/no) yes

If yes, please address the following points:

3.2. A motivation is given for why the experiments are conducted on the selected datasets (yes/partial/no/NA) yes

3.3. All novel datasets introduced in this paper are included in a data appendix (yes/partial/no/NA) NA

3.4. All novel datasets introduced in this paper will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no/NA) NA

3.5. All datasets drawn from the existing literature (potentially including authors' own previously published work) are accompanied by appropriate citations (yes/no/NA) yes

3.6. All datasets drawn from the existing literature (potentially including authors' own previously published work) are publicly available (yes/partial/no/NA) yes

3.7. All datasets that are not publicly available are described in detail, with explanation why publicly available alternatives are not scientifically satisficing (yes/partial/no/NA) NA

4. Computational Experiments

4.1. Does this paper include computational experiments? (yes/no) yes

If yes, please address the following points:

4.2. This paper states the number and range of values tried per (hyper-) parameter during development of the paper, along with the criterion used for selecting the final parameter setting (yes/partial/no/NA) yes

4.3. Any code required for pre-processing data is included in the appendix (yes/partial/no) yes

4.4. All source code required for conducting and analyzing the experiments is included in a code appendix (yes/partial/no) yes

4.5. All source code required for conducting and analyzing the experiments will be made publicly available upon publication of the paper with a license that allows free usage for research purposes (yes/partial/no) yes

4.6. All source code implementing new methods have comments detailing the implementation, with references to the paper where each step comes from (yes/partial/no) yes

4.7. If an algorithm depends on randomness, then the method used for setting seeds is described in a way sufficient to allow replication of results (yes/partial/no/NA) yes

4.8. This paper specifies the computing infrastructure used for running experiments (hardware and software), including GPU/CPU models; amount of memory; operating system; names and versions of relevant software libraries and frameworks (yes/partial/no) yes

4.9. This paper formally describes evaluation metrics used and explains the motivation for choosing these metrics (yes/partial/no) yes

4.10. This paper states the number of algorithm runs used to compute each reported result (yes/no) yes

4.11. Analysis of experiments goes beyond single-dimensional summaries of performance (e.g., average; median) to include measures of variation, confidence, or other distributional information (yes/no) yes

4.12. The significance of any improvement or decrease in performance is judged using appropriate statistical tests (e.g., Wilcoxon signed-rank) (yes/partial/no) yes

4.13. This paper lists all final (hyper-)parameters used for each model/algorithm in the paper's experiments (yes/partial/no/NA) yes